

Module 06: Controlled Access & Channelization Protocols

Unit 2: Data Link Layer & MAC Sublayer

Collision-Free Protocols & CDMA

Comprehensive coverage of collision-free access (Reservation, Polling, Token Passing) and channelization methods (FDMA, TDMA, CDMA) with Walsh matrix mathematical proofs and solved numericals.

Controlled Access & Channelization: Reservation, Polling, Token Passing & CDMA Walsh Codes

1. Controlled Access: Collision-Free Methods

A. Polling (Primary-Secondary Master/Slave):



B. Token Passing (Logical Ring Architecture):



Station holding TOKEN transmits for Token Hold Time (THT).

Guaranteed Zero Collisions & Deterministic Delay!

Trade-off: High overhead of polling/token management at low load.

2. CDMA & Orthogonal Walsh Codes

Walsh Matrix Generation (N = 4):

$W1 = [+1, +1, +1, +1]$ $W2 = [+1, -1, +1, -1]$
 $W3 = [+1, +1, -1, -1]$ $W4 = [+1, -1, -1, +1]$
Property: $W_i \cdot W_i = N = 4$ AND $W_i \cdot W_j = 0$ (Orthogonal!)

Encoding & Transmission:

Data Bit 0 $\rightarrow -1$ | Data Bit 1 $\rightarrow +1$ | Silent $\rightarrow 0$

Transmitted Vector: $S_i = d_i \cdot W_i$

Composite Channel Signal: $C = S1 + S2 + S3 + S4$

Receiver Decoding (Dot Product):

Decoded Data $d_i = (C \cdot W_i) / N$

Channelization Architecture: FDMA vs TDMA vs CDMA Comparison & Solved Example

FDMA: Bandwidth divided into dedicated frequency bands. Requires Guard Bands. Continuous transmission.

TDMA: Entire bandwidth shared in discrete time slots. Requires synchronization & Guard Times. Burst transmission.

CDMA: Stations transmit simultaneously across the entire bandwidth using unique orthogonal codes. No time slots!

AKTU Example:

If S1 sends $d1=+1$, S2 sends $d2=-1$, S3 silent ($d3=0$), S4 sends $d4=+1$:

$C = (+1)W1 + (-1)W2 + (0)W3 + (+1)W4 = [1, 1, 1, 1] + [-1, 1, -1, 1] + [1, -1, -1, 1] = [1, 1, -1, 3]$

1. Controlled Access vs Contention Protocols

Unlike contention methods where stations collide, controlled access guarantees deterministic, collision-free access. Stations either reserve time slots beforehand (Reservation), respond to master inquiries (Polling), or pass a logical permission token (Token Ring/Bus).

2. CDMA & Orthogonal Walsh Codes

Code Division Multiple Access allows all stations to transmit simultaneously across the entire bandwidth. Each station is assigned a unique, mutually orthogonal chip sequence generated via Walsh matrices.

CDMA Encoding & Decoding Formulas

Transmitted Signal: $S_i = d_i \cdot W_i$

Composite Signal: $C = \sum S_i$

Receiver Decoding: $d_i = (C \cdot W_i) / N$